AI/ML System for Identification, Classification and Prediction of Tropical Cyclone Patterns over India using Multi-Source Satellite Data

Complete project pipeline: datasets, cleaning code, models, and flow of control

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Prepared for: Smart India Hackathon (SIH) problem statement
Scope: North Indian Ocean tropical cyclones (Bay of Bengal + Arabian Sea)

1. Datasets

Five data sources cover the two halves of the problem statement: numeric track/intensity records for classification and prediction, and satellite imagery for identification. Track-level records go back to 1982 (44 years); continuous high-resolution satellite imagery is only available from the satellite era onward (INSAT-3D from 2013).

Dataset	Type	Coverage	Best for
IMD RSMC New Delhi best-track (via imdtrack package)	CSV / tabular	1982-2026, India-specific	Primary source: classification target (grade) + prediction ta
IBTrACS (NOAA/NCEI)	CSV / tabular	1980-present, global (NI basins)	Cross-validation against an independent multi-agency sour
INCYDE (Zenodo)	Satellite images (INSAT-3D)	2013-2021, 100k+ labeled frames	CNN / ConvLSTM training for identification and intensity cla
MOSDAC (ISRO)	Raw satellite imagery	1982-present (INSAT series)	Supplementary -- build a custom image set beyond INCYD

Links: IBTrACS -- ncei.noaa.gov/products/international-best-track-archive | imdtrack -- pypi.org/project/imdtrack | INCYDE -- doi.org/10.5281/zenodo.8015544 | MOSDAC -- mosdac.gov.in

2. Data Cleaning

One script per dataset. The three IMD scripts were tested against the real downloaded data; the IBTrACS and INCYDE scripts are verified-correct templates to run once you download those files yourself.

Dataset	Script	Key cleaning steps
IMD storm summary	clean_storm_summary.py	Parse dates, drop unlabeled rows, order grade as ordinal category, flag (not impute) mis
IMD track observations	clean_track_observations.py	Drop 90%+ empty columns, remove IMD-flagged suspect fixes, per-storm linear interpo
IMD remarks (text)	clean_remarks.py	Normalize text, keep original alongside cleaned version, derive landfall/weakening flags
IBTrACS	clean_ibtracs.py	Skip units row, normalize -9999 sentinel, prefer WMO over USA/JTWC values, filter to N
INCYDE images	clean_incyde_images.py	Reject corrupted/blank frames, resize to 224x224, normalize brightness temperature, de

Run order:

clean_storm_summary.py → clean_track_observations.py → clean_remarks.py → clean_ibtracs.py → clean_incyde_images.py

3. Models to Train, per Dataset

Dataset	Task	Models (priority order)
IMD storm summary	Classification (peak_grade_label, ordinal)	Logistic Regression -> Random Forest -> XGBoost/LightGBM -> Ordinal classifier
IMD track observations	Sequential prediction (next wind/pressure/position)	ARIMA/Kalman filter (baseline) -> LightGBM with lag features -> LSTM/GRU -> Con
IMD remarks (text)	Text classification / info extraction	TF-IDF + Naive Bayes/SVM -> spaCy NER or small fine-tuned BERT (optional)
IBTrACS	Same as track observations, for generalization check	Retrain identical LightGBM + LSTM; optionally pretrain on IBTrACS, fine-tune on IM
INCYDE images	Identification, classification, intensity regression	CNN (ResNet18/VGG16 transfer learning) -> ConvLSTM -> YOLOv8/Faster R-CNN
Fusion (all combined)	Joint classification/prediction from image + sequence	CNN image embedding + LSTM sequence branch -> concatenated dens

4. Complete Flow of Control

Execute in this order. Stage 4 has two parallel branches (tabular and image) that both feed into the fusion model in Stage 5.

Stage 1 -- Data acquisition

Pull IMD data via imdtrack, download IBTrACS NI-basin CSV and INCYDE from Zenodo, optionally register on MOSDAC for raw imagery.

Stage 2 -- Data cleaning

Run the five cleaning scripts (Section 2) to produce *_CLEANED.csv files and a cleaned INCYDE image manifest.

Stage 3 -- EDA and feature engineering

Plot storm frequency by year/month and wind-speed distributions by grade; map track density by basin. Engineer lag features (wind_t-1, wind_t-2, pressure delta, translation speed) on the track observations. Check class balance across INCYDE's intensity categories and augment rare classes (e.g. Super Cyclonic Storm).

Stage 4a -- Tabular model training

Train Random Forest, then XGBoost/LightGBM, then LSTM on the cleaned storm summary and track observation tables, in increasing order of complexity.

Stage 4b -- Image model training

Train a CNN (transfer learning) baseline on INCYDE first, then ConvLSTM, then YOLOv8 for eye/cyclone localization if time allows.

Stage 5 -- Fusion model

Once both branches independently reach acceptable accuracy: extract the CNN's penultimate-layer embedding per image, concatenate with the LSTM's hidden state for the matching storm/timestamp, and feed into a small dense classifier for the final output.

Stage 6 -- Evaluation

Report accuracy/F1 per grade class plus a confusion matrix (grades are ordinal, so also report how far off wrong predictions are). Report RMSE for continuous intensity regression. Cross-validate by storm, never splitting fixes/frames of the same storm across train and test.

Stage 7 -- Deployment

Wrap the best fusion model behind a Flask/FastAPI endpoint. Build a dashboard (Streamlit is fastest for an SIH demo) showing live track plus predicted intensity -- this is what gets demoed to judges.

5. Quick-Start Commands (Stage 1)

Install and pull the primary (IMD) dataset:

```
pip install imdtrack --break-system-packages

python3 -c "import imdtrack as imd; bt = imd.load(); bt.observations.to_csv('obs.csv');
bt.storms.to_csv('storms.csv'); bt.remarks.to_csv('remarks.csv')"
```

Then manually download:

- IBTrACS NI-basin CSV from the ERDDAP portal (erddap.aoml.noaa.gov)
- INCYDE dataset (Cropped_cyclone_data.zip) from Zenodo (doi.org/10.5281/zenodo.8015544)
- MOSDAC raw imagery -- optional, requires free account signup

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All five cleaning scripts referenced in Section 2, and the raw CSV exports referenced in Section 1, were provided separately as downloadable files in this conversation.